Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20250811-029-DS-4-RA-101	Issued	2025-08-15

UT Thickness Data Sheet - 4-RA-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 4-RA-101 located within the ATU-100. The scope involves the 6-month re-measurement of the line to monitor wall loss and validate the current fitness-for-service assumptions.

Marcus Chen, acting as the lead UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., performed the measurements and data recording. All activities were executed in accordance with the requirements defined in QSCP-NDT-001 Rev 4.

Technical Specifications

The subject line, 4-RA-101, is constructed from A106 Gr.B with a nominal wall thickness of 8.56 mm and an outer diameter of 114.3 mm. The design limits for this rich amine service are 862 kPa(g) and 82 °C.

The minimum allowable thickness (t-min) for this line is 3.81 mm. All thickness measurements were conducted following the guidelines of ASTM E797.

Measurement Results

Thickness data was collected as part of the 6-month re-measurement detailed in LNDE-RPT-20250811-029-R0. The measurements confirm that the corrosion rate remains within the FFS assumption for most of the line.

At CML 4-RA-101-06, the recorded wall thickness is 3.3 mm. This value is below the t-min of 3.81 mm, and the remaining life is listed as expired.

Analysis and Compliance

Per site rules, the reading of 3.3 mm at CML 4-RA-101-06 is below the retirement thickness, necessitating the issuance of an NCR.

The remaining life calculation for this point is based on the larger of the long-term and short-term corrosion rates. Because the thickness has fallen below 3.81 mm, the asset integrity of this section is compromised.

Inspection Interval

Line 4-RA-101 is classified as rich amine service, which carries a corporate RBI interval of 3 years.

Due to the expired remaining life at CML 4-RA-101-06, the standard inspection frequency is superseded by the requirement to address the non-conformance.

Thickness readings, measured 2025-08-15

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2024-11-04	6.50	6.32	0.075	0.166	15.1	2028-08-15	
4-RA-101-02	8.56	3.81	2024-11-04	6.46	6.40	0.073	0.170	15.2	2028-08-15	
4-RA-101-03	8.56	3.81	2024-11-04	5.52	5.42	0.108	0.209	7.7	2028-08-15	gauge photo attached
4-RA-101-04	8.56	3.81	2024-11-04	6.90	6.82	0.061	0.124	24.4	2028-08-15	
4-RA-101-05	8.56	3.81	2024-11-04	6.37	6.20	0.082	0.168	14.2	2028-08-15	
4-RA-101-06	8.56	3.81	2024-12-10	3.55	3.30	0.423	0.625	expired	2026-02-13	< t-min; gauge photo attached
4-RA-101-07	8.56	3.81	2024-11-04	7.21	7.10	0.051	0.100	33.0	2028-08-15	
4-RA-101-08	8.56	3.81	2024-11-04	5.06	4.85	0.130	0.275	3.8	2027-07-07	gauge photo attached
4-RA-101-09	8.56	3.81	2024-11-04	7.17	7.04	0.052	0.110	29.4	2028-08-15	
4-RA-101-10	8.56	3.81	2024-11-04	6.91	6.86	0.061	0.118	25.8	2028-08-15	
4-RA-101-11	8.56	3.81	2024-11-04	5.53	5.34	0.113	0.227	6.8	2028-08-15	gauge photo attached
4-RA-101-12	8.56	3.81	2024-11-04	6.10	5.95	0.185	0.185	11.5	2028-08-15	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II	2025-08-15	_____